Computer Networks

12th November 2024

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions

Marks

Question 1

Explain the TCP/IP model by answering the following:

a) Describe the primary function of the Internet Layer. **2**

Mark Scheme

The Internet Layer is responsible for addressing, routing, and forwarding packets across networks to their destinations (1 mark). It determines the best path for data to travel from the source to the destination (1 mark).

- ***1 mark for mentioning device addressing and routing.***
- ***1 mark for explaining the determination of the best path for data.***

b) Identify a device that operates at the Network Access Layer **and** explain its role. **2**

Mark Scheme

A switch (1 mark) operates at the Network Access Layer. Its role is to forward data frames between devices on the same network, using MAC addresses to determine the destination of each frame (1 mark).

- ***1 mark for identifying a switch.***
- ***1 mark for explaining its role in forwarding data frames using MAC addresses.***
- ***Award marks for Bridge (even though it is obsolete)***

c) Explain the primary function of Transport Layer in details (including the handshaking procedure) **and** describe how session-related functions such as establishing, maintaining and terminating communications are handled in TCP/IP model. **4**

Mark Scheme

The Transport Layer ensures reliable data transfer, error detection, and recovery, providing end-to-end communication services. The transport layer handshaking procedure, commonly seen in TCP (Transmission Control Protocol), involves a three-step process known as the three-way handshake, where the client sends a synchronization (SYN) packet to the server, the server replies with a SYN-ACK (synchronization acknowledgment), and the client responds with an ACK to establish a reliable connection. This exchange ensures that both parties are ready to send and receive data, providing a stable connection before data

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transmission begins. (2 mark). In TCP/IP model, session management functions are distributed between application and transport layer, with the application layer handling most session initiation, management and termination (2 mark).

- ***2 marks for describing the Transport Layer's function.***
- ***2 marks for describing the Session Layer's function handling in TCP/IP.***

d) Identify the layer of the TCP/IP model responsible for data encryption and decryption. Provide an example of a protocol that operates at this layer. **2**

Mark Scheme

The Application Layer in TCP/IP model is the main layer responsible for data encryption and decryption. (1 mark) An example of a protocol that operates at this layer is SSL/TLS (Secure Sockets Layer/Transport Layer Security) (1 mark)

- ***1 mark for correctly identifying the Application Layer and its role as the presentation layer in TCP/IP.***
- ***1 mark for providing an example of a protocol (SSL/TLS).***

Total 10 marks

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Question 2

a) What is the message encapsulation process **and** what is the data called in each OSI layer? **8**

Mark Scheme

Message encapsulation refers to the process of wrapping data with the necessary protocol information as it moves down through the layers of OSI model, so each layer adds its own data (known as header) to the data passed from the above layer. The data can be the controls or addressing information related to each layer.

Data Names at Each OSI Layer

1. ***Application Layer: Data***
2. ***Presentation Layer: Data***
3. ***Session Layer: Data***
4. ***Transport Layer: Segment (TCP) / Datagram (UDP)***
5. ***Network Layer: Packet***
6. ***Data Link Layer: Frame***
7. ***Physical Layer: Bits***

1- Explanation of Message Encapsulation (1 mark)
2- Identification of data name in each OSI layers (7 marks in total, 1 mark is allocated to identification of data name in each layer)

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b) Given the following scenario, identify **and** justify the most suitable types of networks to be used (LAN, WAN, MAN) **2**

A global corporation with offices in multiple cities in the UK needs to connect its regional data centres to facilitate secure data transfer and to communicate. The corporation also needs computers, printers and server connections for TWENTY (20) employees within each single office building for file sharing and communication.

Mark Scheme

A LAN (Local Area Network) is a network that connects computers and devices within a small area such as home, office building or campus. LAN is the appropriate network type for the corporate to connect devices in each office.

A WAN (Wide Area Network) covers a large area, connecting multiple LANs over long distances, such as cities or countries. WAN is the appropriate network type to connect offices in multiple cities.

1. ***Correct Identification and justification of WAN network to connect the corporate's data centres in multiple countries (1 mark)***
2. ***Correct Identification and justification of LAN network to connects computer devices in each office building (1 mark)***

Total 10 marks

Question 3

You are managing a small company's LAN with TWENTY (20) computers in a star topology using a single switch. The company plans to expand, adding FORTY (40) more employees and requires wireless connectivity for mobile devices, laptops and visitors. Your task is to redesign the network into a tree topology and integrate wireless access points to accommodate both wired and wireless devices. Ensure the network is scalable, efficient and secure.

a) Explain **THREE (3)** advantages of the tree topology over star topology. **6**

Marking scheme

Three advantages of tree topology over star topology are: scalability in which tree topology is a more scalable extension to the LAN. (1 mark) More nodes and segments can easily be added by attaching new subnets (1 mark). Secondly, in tree topology faults and troubleshooting is easier.(1 mark) If a segment fails, it does not affect the rest of network (1 mark). Thirdly, there is less network congestion(1 mark) and better network traffic in tree topology, since network is segmented more efficiently in tree compared to star topology (1 mark).

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b) Identify the most suitable cabling based on bandwidth needs, distance limits and scalability for high-speed data transfer. Justify your choice. **2**

Mark Scheme

The best option for cabling is Cat 6a Ethernet cable which can support up to 10 Gbps over a 100-metre distance (1 mark), which should be sufficient for the network in this scenario to connect workstations, access points and other devices regarding their distances (1 mark).

c) Recommend the best Wi-Fi standard to handle device load, optimise speed, reduce interference and ensure secure access for all users. Justify your choice. **2**

Mark scheme

This should also accommodate the additional floors and adding up more workstations in the future (1). The bandwidth is also ample for high-speed data transfer required by the growing network (1 mark).

Total 10 marks

Question 4

a) Explain what a security attack is. **1**

Mark Scheme

A security attack is any action that compromises the security of information or systems (1 mark).

b) For each of the following incidents, identify **and** explain the type of attack. Additionally, recommend **TWO (2)** security countermeasures for each incident to prevent similar threats in the future.

i) **Incident 1**: An attacker intercepted sensitive customer data, including personal and financial details, while it was being transmitted over the network. **3**

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Mark scheme

Incident 1: This is an eavesdropping attack. In this attack the attacker intercepts communication between two parties without their knowledge (1 mark).

Countermeasures:

- ***implement strong encryption protocols (1 mark)***
- ***use IDS/IPS could also help to detect and prevent similar attacks (1 mark)***

ii) **Incident 2:** An organised attack overwhelmed the company's web server, making it unavailable to customers for several hours. **3**

Mark scheme

Incident 2: The attack is a type of DDOS attack (Distributed Denial of Service), where multiple compromised systems, often part of botnet, flood a target such as a (web) server or network with excessive traffic with the aim to overwhelm its resources, causing it to become slow or completely shut down (1 mark).

Countermeasures:

- ***DDOS Protection Services can detect and filter out malicious traffic before it reaches the serve (1 mark).***
- ***Rate Limiting and Traffic Analysis can control the number of requests a server can handle and use traffic analysis tools to identify and block suspicious traffic patterns (1 mark).***

iii) **Incident 3:** An employee inadvertently downloaded an infected email attachment, which subsequently locked the system and demanded a payment for decryption. **3**

Mark scheme

Incident 3: Ransom attack is a type of malware that encrypts the victim's files and demands a ransom payment for the decryption key. This often occurs through phishing emails with malicious attachments (1 mark).

Countermeasures:

- ***Employee Training: Conduct regular cybersecurity awareness training to educate employees about the dangers of phishing and how to recognise suspicious emails (1 mark).***
- ***Anti-Malware Solutions: Deploy advanced anti-malware and endpoint protection solutions to detect and block ransomware before it can execute (1 mark).***

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- **Backup solutions (alternative student answer): corporate can have a backup copy of the information or services in the organisation (real or virtual back-up), so in case the system is locked, the backup can serve.**

Total 10 marks

Question 5

a) This question is about firewall and intrusion detection system (IDS).

i) Define a firewall **and** explain its primary purpose in network security. **2**

Mark Scheme

A firewall is a network security device that monitors and controls incoming and outgoing network traffic based on predetermined security rules (1 mark). Its primary purpose is to establish a barrier between a trusted internal network and untrusted external networks, such as the internet, to prevent unauthorized access (1 mark).

- ***1 mark for defining a firewall.***
- ***1 mark for explaining its primary purpose.***

ii) Define an Intrusion Detection System (IDS) **and** explain its primary purpose in network security. **2**

Mark Scheme

An Intrusion Detection System (IDS) is a device or software application that monitors a network or systems for malicious activity or policy violations (1 mark). Its primary purpose is to detect and alert administrators to potential security breaches or attacks (1 mark).

- ***1 mark for defining an IDS.***
- ***1 mark for explaining its primary purpose.***

iii) Compare **and** contrast the roles of firewalls **and** IDS in network security. **2**

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Mark Scheme

Firewalls prevent unauthorised access (1 mark), IDS detects and reports on potential security incidents (1 mark).

- ***1 mark each contrast highlighted.***

b) List **and** briefly explain **TWO (2)** types of firewalls in regard to their functionality. **4**

Mark Scheme
Packet-Filtering Firewall (1 mark): Inspects packets and allows or blocks them based on source and destination IP addresses, ports, or protocols (1 mark).

Stateful Inspection Firewall (1 mark): Tracks the state of active connections and makes decisions based on the context of the traffic, providing more advanced filtering than packet-filtering firewalls (1 mark).
Application gateway, proxy serve also valid.

- ***1 mark for listing and 1 mark for explaining each type of firewall.***

Total 10 marks

Question 6

You are the network administrator for a medium-sized company with approximately 150 employees operating a local area network (LAN). The company requires secure remote access for employees who need to connect to the company network from outside locations, such as their homes or while travelling.

a) Identify, explain, **and** provide **ONE (1)** example of the types of network software required for this network, including the appropriate system software for servers, client machines **and** network devices. **6**

Mark Scheme

Servers Software:

1. ***operation system (1 mark): (OS) is the fundamental software required for LAN servers. Options include Windows or Linux OS (*Ubuntu Server** or **CentOS*) (1 mark).***

Client Machines Software:

1. ***Web Browsers allow users to access web-based resources (1 mark). Example Google Chrome (1 mark)***

Network Devices Software:

1. ***Network Management Software help in monitoring, managing, and troubleshooting network performance (1 mark). Example: SolarWinds Network Performance Monitor (1 mark).***

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b) Identify the most suitable option and types for the company's LAN remote connection. **1**

Mark Scheme

System for remote connection (1 mark)

VPN or Secure VPN (SSL VPN) are the best options for remote access to the network.

c) Explain THREE (3) fundamental functions of the software identified in b). **3**

Mark Scheme

Key functionalities of VPN include:

- ***Authentication: to validate the data is sent by an authorised remote user***
- ***Access control: to prevent unauthorised remote access to the network resources***
- ***Confidentiality: using encryption to prevent data from being read or copied when it is being transferred from remote user***
- ***Integrity: to ensure the data has not been amended in the transmission***

1 mark for each bullet point to a max of 3 marks.

Total 10 marks

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Question 7

List **and** explain **FIVE (5)** issues of VOIP services noting the problem and their cause. **10**

Mark Scheme

Any five issues with VOIP services along with their courses from below options are acceptable.

1. ***Latency occurs when the voice or video packets take longer time to arrive to destination (delay between when a speaker says something and when the listener hears it on a VoIP call). This issue can be caused by packet switching overhead (the process of dividing data into smaller packets and transferring them through different routes to destination and packets reassembly at the destination), high amount of traffic on the network, long physical distance, insufficient bandwidth, etc.***
2. ***Jitter is also the delay experienced in receiving a packet when a packet is expected to arrive at a certain time. When sending a sequence of packets from point A to point B, each packet may take slightly different times to reach the destination due to varying network conditions. This variation, known as jitter, isn't a problem for tasks like downloading web pages but can impact on real-time data transmission, such as voice or video. In real-time applications, consistent packet delivery is crucial to maintain smooth communication and avoid issues like delays, echo or degraded quality.***
3. ***Bandwidth is the maximum amount of data that can be transmitted over a network connection in a given amount of time, typically measured in bits per second (bps). This amount is shared between voice and other computer data in a communication. So having loads of data transfer or multiple VOIP calls may decrease the bandwidth allocation to each communication and can cause issues in VOIP communication.***
4. ***Packet loss: packet loss in network communication is unavoidable, especially in UDP based communications such as VOIP, since error correction is not included. Packet loss on VoIP is similar to when a call on a cell/mobile phone begins to "break-up". This results in the receiver missing parts of the conversation.***
5. ***Reliability: A reliable network connection is essential for VoIP to maintain continuous, high-quality communication. any instability in network including bandwidth, lost packets, no redundancy, etc. can disrupt the voice communication***
6. ***Security: VOIP calls may also be target of cyber-attacks such as eavesdropping, Man-in-the-Middle, DOS, packet sniffing etc. These may violate the confidentiality, integrity and availability of the communication and VOIP service.***
7. ***Scalability: Above listed issues can cause difficulty in VOIP to extend the capacity and add more telephony conversations in a business.***
8. ***Interoperability: VoIP equipment and software must be harmonised and standardised across different manufacturers to ensure compatibility. Without standardisation, inconsistencies between vendors' VoIP systems can prevent seamless communication, making it difficult or impossible for users on different platforms to connect reliably.***

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1- identification and explanation of the problem for 5 issues (5 marks in total, 1 mark for identification and explanation of each issue)
2- explanation of the causes of each of the five identified issues (5 marks in total, 1 mark for explanation of the cause/s of each identified issue)

Total 10 marks

Question 8

You are tasked with planning the IP address allocation for the following network:

Small Office Network: A small business with **25 devices** (computers, printers, and mobile devices) that needs a private local area network (LAN) for file sharing, printing and internet access. This business has 3 separate departments.

a) Specify the **IP v4 address class** that would be most suitable for the Small Office Network. Justify your selection with regards to nodes allocation. **2**
Marking Scheme

For the Small Office Network with 25 devices, the most suitable IP address class would be Class C (1 mark). Class C addresses allow for up to 254 usable IP addresses within a single subnet (from .1 to .254, excluding the network and broadcast addresses). This is more than enough to accommodate the 25 devices in the office, which includes computers, printers, and mobile devices (1 mark).

b) Give example IP address range for each network (network segments) **4**

Mark scheme

Given that Class C IP addresses range from 192.0.0.0 to 223.255.255.255, we can choose a private IP range within this class. A common private IP range for Class C is 192.168.0.0 to 192.168.255.255.

Let's allocate the IP addresses as follows:

- ***Network Address range: 192.168.1.0/24 to 192.168.255.255***
- ***Subnet 1 (Department 1): 192.168.1.1/24 to 192.168.1.25 (supports up to 254 devices)***
- ***Subnet 2 (Department 2): 192.168.2.1/24 to 192.168.2.25(supports up to 154 devices)***
- ***Subnet 3 (Department 3): 192.168.3.1/24 to 192.168.3.25 (supports up to 254 devices)***

1 mark for each bullet point to max of 4 marks.

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c) Explain why you chose that specific IP class for each network, considering factors such as: **4**

- Number of devices supported
- Future expansion
- Segmentation requirements
- Private vs. public IP addresses

Mark scheme

Explanation for Choosing Class C

- ***Class C networks can support up to 254 devices, which is more than sufficient for the 25 devices in the small office network.***
- ***Each department can have its own segment, reducing broadcast traffic and improving performance.***
- ***Using private IP addresses (192.168.x.x) ensures that the network is secure and not directly accessible from the internet. This is crucial for a small office network to maintain security and privacy.***
- ***Class C IP addressing is commonly used in small and internal private LANs and not in public networks. This means they can freely be used in internal domain of businesses and will not conflict with public IP addresses. NAT (Network Address translation) scheme is used to translate private to public addresses in case LAN needs connection to internet and external networks.***

1 mark for each bullet point to max of 5 marks.

Total 10 marks

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Question 9

You are the network administrator for a medium-sized office building with multiple floors. The building has an open-plan layout on each floor, along with several enclosed meeting rooms. The management has requested the implementation of a Wireless Local Area Network (WLAN) to support approximately 100 devices, including laptops, smartphones, and tablets. However, during your initial site survey, you observe some challenges:

1. You notice that certain areas of the building, especially the far corners of each floor and enclosed rooms, experience weak or no signal coverage.
2. Thick concrete walls that separate rooms and floors.
3. Numerous electronic devices, such as microwaves and cordless phones, that operate on the frequency bands 2.4 GHz and 5 GHz
4. Nearby wireless networks from neighbouring buildings that may cause overlapping channels and congestion.
5. You notice an issue of neighbouring building employees connecting to the guest user account of the Wi-Fi.

a) Explain how each of the FIVE (5) challenges identified above can impact effective Wi-Fi implementation. **5**

Mark scheme

The challenges observed in this scenario are generally a range or interference issues that may be evidenced in wireless implementation.

Problem in (1), is mainly an indoor range issue which is caused by building design and room layout. the WLAN standard can support a range of up to 300 metres in ideal conditions. However, if the building design and layout of room extend behind this limitation, wireless signals become weak and cannot reach destination devices far from access points.

Problem in (2), is again an indoor issue caused because of the construction materials, mainly the thick walls. Thick walls, particularly those made of materials like concrete, brick, or metal, can weaken the radio waves used in wireless communication. It can also be a multipath interference issue caused when transmitted signals reach the receiving devices through multiple paths due to obstacles in the environment, (the wall).

Problem in (3), is an interference issue caused by the signals reflected from other electronical equipment. This is called an external interference which is caused by signals broadcast from devices with same frequency.

Problem in (4), is also an interference issue caused by the signals reflected by other wireless networks.

Problem in (5) is a security issue caused by the weak password setting policy for the guess user account.

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b) Identify **ONE (1)** strategy to resolve each of the challenges you have observed in (a). **5**

Mark scheme

Problem in (1) may be resolved by adding more access points in the building to cover all devices in various dimensions and distances, or using access points with stronger antenna, or using repeaters.

Problem in (2) may be resolved by moving access points closer to the wall or mounting them on wall, also by using AP with stronger antenna.

Problem in (3) may be resolved by isolating the microwave and cordless phones to areas out of wireless range.

Problem in (4) may be resolved by conducting site survey, identifying channel used by neighbours and manually configuring AP to another channel. A simple solution to be tried is also to move the AP to windows.

Problem in (5) can be resolved by using strong password policy. Enabling WPA3 could also help to resolve this issue.

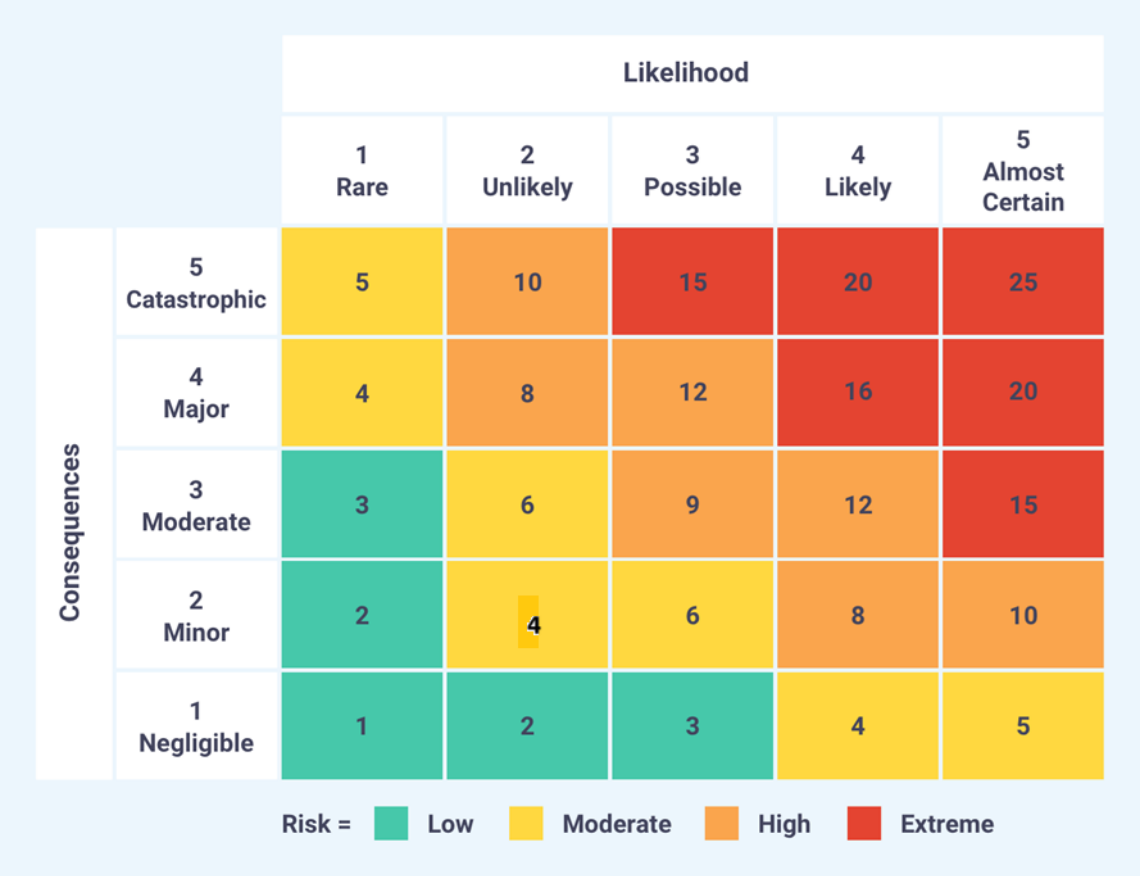
- ***Identification and explanation of the issue type and sub-category (5 marks in total, 1 mark allocated to identification and explanation of the issue type for each challenge as listed in the question)***
- ***Identification of one strategy to resolve the challenges (5 marks in total, 1 mark allocated to identification of one strategy for each challenge)***

Total 10 marks

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Question 10

Below is a Risk Assessment Matrix outlining various risks associated with a company's network infrastructure. **10**



Using the provided Risk Assessment Matrix, try to fill out the following risk assessment spreadsheet by identifying the elements should they be appropriate to each blank cell in the table (2(a), 2(b), etc).

Serial	Asset	Threat	Vulnerability	Loss of CIA	Severity	Likelihood	Risk (1-25)	Action	Control measure
1	Web server	flood	Water damage	A	5	1	5	mitigate	backup
2	Web server	theft	2(a)	2(b)	5	2	2(c)	2(d)	2(e)
3	Web server	Packet sniffing	3(a)	3(b)	4	4	3(c)	3(d)	3(e)

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Mark Scheme

Serial	Asset	Threat	Vulnerability	Loss of CIA	Severity	Likelihood	Risk (1-25)	Action	Control measure
1	Web server	flood	Water damage	A	5	1	5	mitigate	backup
2	Web server	theft	***Physical security breach***	***C***	5	2	***10***	***mitigate***	***Physical security measures, encryption***
3	Web server	Packet sniffing	***Insecure network***	***C***	4	4	***16***	***mitigate***	***Use encryption (e.g., SSL/TLS), secure network configurations***

1 mark for each cell with correct answer.

Total 10 marks

END OF PAPER

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Learning Outcomes matrix

Question	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	1	Yes
2	1,2	Yes
3	2,4	Yes
4	3	Yes
5	3	Yes
6	5,6	Yes
7	6	Yes
8	1	Yes
9	4	Yes
10	3	Yes

Grade Descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Understand network and communication protocols	Has little to no awareness of principles and concepts underlying of network and approaches and is unable to identify associated protocols.	Has vague awareness of principles and concepts underlying of network and approaches and is only superficially able to identify associated protocols.	Has satisfactory awareness of principles and concepts underlying theoretical frameworks and approaches and demonstrates sufficient ability to identify associated protocols.	Has very good awareness of principles and concepts underlying frameworks and approaches and is able in detail to identify associated protocols.	Has an excellent awareness of principles and concepts underlying network and approaches and is comprehensively able to identify associated protocols.
Understand the principles of common network topologies and architectures	Has little to no awareness of principles and concepts underlying network topologies and approaches and is unable to identify associated architecture.	Has vague awareness of principles and concepts underlying network topologies and approaches and is only superficially able to identify associated architecture.	Has satisfactory awareness of principles and concepts underlying network topologies and approaches and demonstrates sufficient ability to identify associated architecture.	Has very good awareness of principles and concepts underlying network topologies and approaches and is able in detail to identify associated architecture.	Has an excellent awareness of principles and concepts underlying network topologies and approaches and is comprehensively able to identify associated architecture.

Understand how to design and develop a network system	Demonstrates little to no ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an extremely limited ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a satisfactory ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a very good ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an excellent ability to use design principles to effectively develop an artefact to solve an identified issue.
Understand the application of network security measures	Demonstrates little to no ability to adequately use security principles to effectively develop an artefact to solve an identified issue.	Demonstrates an extremely limited ability to adequately use security principles to effectively develop an artefact to solve an identified issue.	Demonstrates a satisfactory ability to use security principles to effectively develop an artefact to solve an identified issue.	Demonstrates a very good ability to use security principles to effectively develop an artefact to solve an identified issue.	Demonstrates an excellent ability to use security principles to effectively develop an artefact to solve an identified issue.
Be able to select and configure the hardware components of a computer network to meet the requirements of a precise specification	Demonstrates little to no ability to adequately develop a network using hardware components.	Demonstrates an extremely limited ability to adequately develop a network using hardware components.	Demonstrates a satisfactory ability to develop a network using hardware components.	Demonstrates a very good ability to develop a network using hardware components.	Demonstrates an excellent ability to develop a network using hardware components.
Be able to define and configure internet telephony and communication systems	Demonstrates little to no ability to adequately define and configure internet telephony and communication systems.	Demonstrates an extremely limited ability to adequately define and configure internet telephony and communication systems.	Demonstrates a satisfactory ability to define and configure internet telephony and communication systems.	Demonstrates a very good ability to define and configure internet telephony and communication systems.	Demonstrates an excellent ability to define and configure internet telephony and communication systems.

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